EE2021 -

Materials Science for Electrical Engineers

Mechanical Properties

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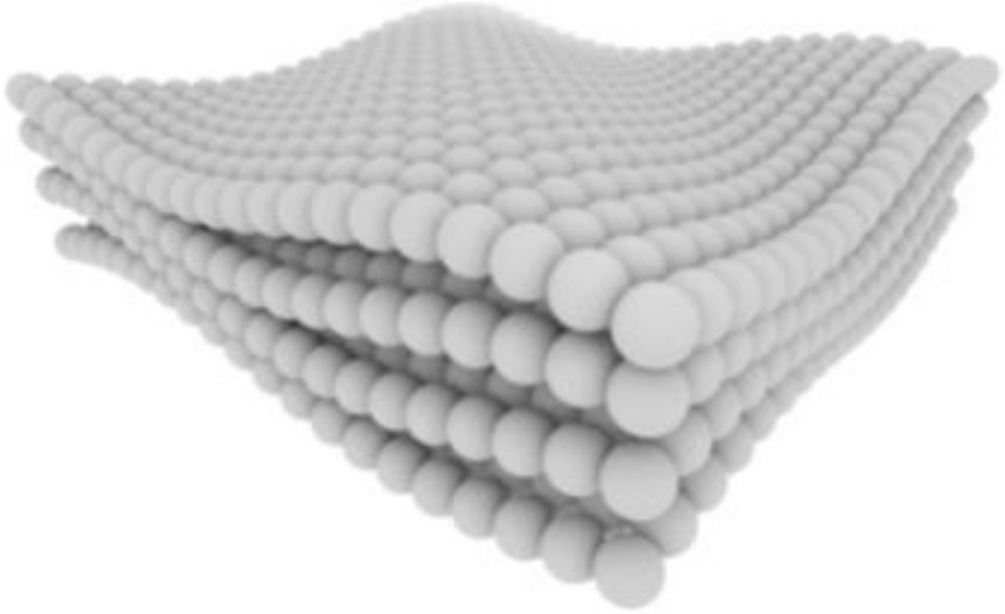
Food for thought

When the automobile industry is moving from internal combustion engine to electric drives, why should an Electrical engineer be concerned about mechanical properties?

Think over the weekend and bring your responses on Monday

What are phonons?

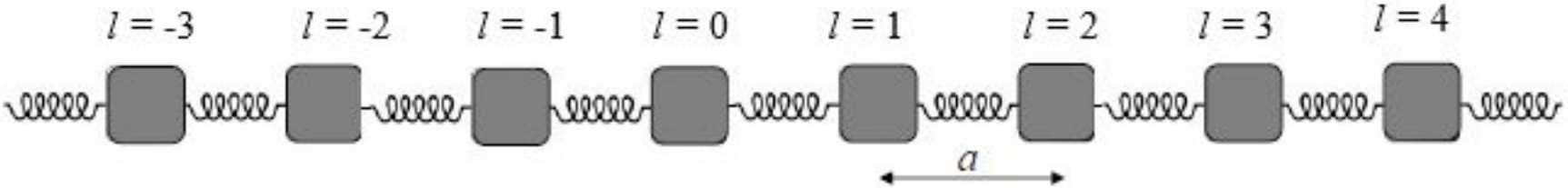
- Phonon is quantized vibration of atoms in a solid — basically, a quantum of lattice vibration energy
- Just as photons are quanta of light (EM waves), phonons are quanta of vibrational waves in a crystal lattice



Sean Kelley via NIST

- Think of atoms in a solid as masses connected by springs
- When they vibrate, waves of displacement travel through the lattice — these are lattice waves
- When we describe these waves quantum mechanically, their discrete energy packets are phonons. Phonons are bosons.
- There are **two main types** of phonons: **Acoustic phonons & Optical phonons**

Monoatomic Linear Chain



- Assumptions- identical atoms of mass M, separated by lattice constant a, quadratic interatomic potential
- Each atom connected to its nearest neighbors with spring constant K
- $u_n(t)$ is **displacement** of the nth atom from its equilibrium position

Equation of motion (1): $$M\frac{d^2u_n}{dt^2} = K[(u_{n+1} - u_n) + (u_{n-1} - u_n)]$$

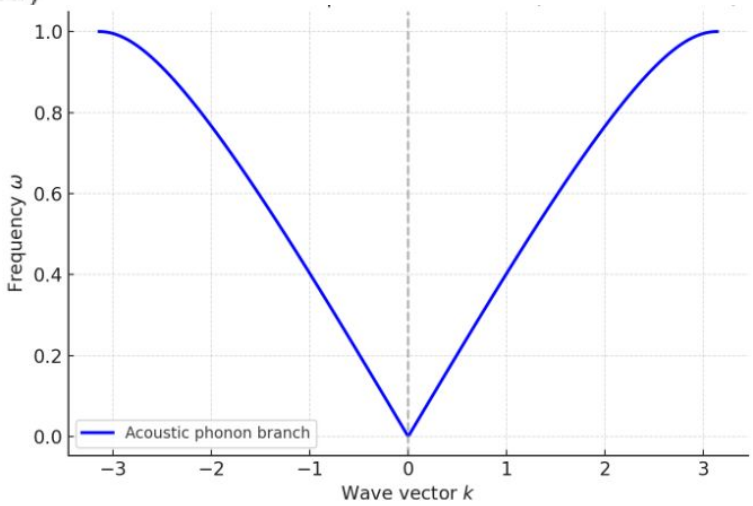
Substituting a traveling wave solution: $$u_n = u_0 e^{i(\omega t - kna)}$$

$$-M\omega^2 u_0 e^{i(\omega t - kna)} = K\left(e^{-ika} + e^{ika} - 2\right) u_0 e^{i(\omega t - kna)}$$

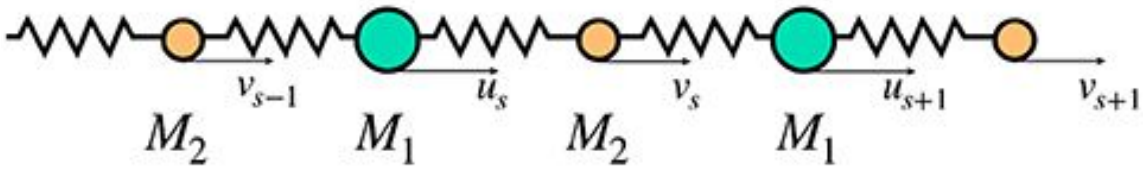
$$M\omega^2 = 2K(1 - \cos(ka))$$

We know, $1 - \cos x = 2\sin^2(x/2)$

Dispersion relation: $$\omega = 2\sqrt{\frac{K}{M}}\left|\sin\left(\frac{ka}{2}\right)\right|$$



Diatomic Linear Chain



- Atoms with mass $\mathbf{M_1}$ & $\mathbf{M_2}$, separated by lattice constant **a**
- Each atom connected to its nearest neighbors with spring constant **K**
- $\mathbf{u_n}$ is displacement for atom 1 and $\mathbf{v_n}$ is for atom 2

Equation of motion (1): $$M_1 \frac{d^2 u_n}{dt^2} = K(v_n - u_n) + K(v_{n-1} - u_n)$$

Equation of motion (2): $$M_2 \frac{d^2 v_n}{dt^2} = K(u_n - v_n) + K(u_{n+1} - v_n)$$

Assume a traveling wave solution: $u_n = u_0 e^{i(\omega t - kna)}, \quad v_n = v_0 e^{i(\omega t - kna)}$

Substituting:

$$-M_1\omega^2 u_0 = K[(v_0 - u_0) + (v_0 e^{-ika} - u_0)]$$

$$-M_2\omega^2 v_0 = K[(u_0 - v_0) + (u_0 e^{ika} - v_0)]$$

Simplifying:

$$-M_1\omega^2 u_0 = K[v_0(1 + e^{-ika}) - 2u_0]$$

$$-M_2\omega^2 v_0 = K[u_0(1 + e^{ika}) - 2v_0]$$

Diatomic Linear Chain

$$\begin{bmatrix} 2K - M_1\omega^2 & -K(1+e^{-ika}) \\ -K(1+e^{ika}) & 2K - M_2\omega^2 \end{bmatrix} \begin{bmatrix} u_0 \\ v_0 \end{bmatrix} = 0$$

For non-trivial solutions, **determinant = 0**

$$(2K - M_1\omega^2)(2K - M_2\omega^2) - K^2(1+e^{ika})(1+e^{-ika}) = 0$$

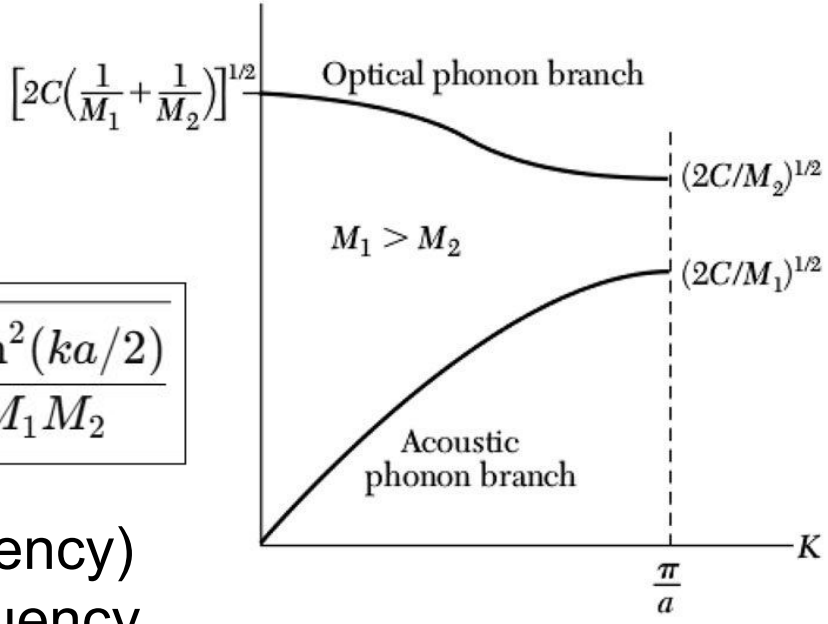
We know, $(1+e^{ika})(1+e^{-ika}) = 2(1+\cos ka) = 4\cos^2(ka/2)$

$$(2K - M_1\omega^2)(2K - M_2\omega^2) - 4K^2\cos^2(ka/2) = 0$$

$$M_1M_2\omega^4 - 2K(M_1+M_2)\omega^2 + 4K^2\sin^2(ka/2) = 0$$

Dispersion relation:

$$\omega^2 = K\left(\frac{1}{M_1} + \frac{1}{M_2}\right) \pm K\sqrt{\left(\frac{1}{M_1} + \frac{1}{M_2}\right)^2 - \frac{4\sin^2(ka/2)}{M_1M_2}}$$



"+" sign → Optical branch (high frequency)
"–" sign → Acoustic branch (low frequency

Acoustic phonons

Nature: Atoms move **in phase** with each other — they all move in the same direction, almost like sound waves in air

Energy: Low energy (long wavelength, small frequency)

Velocity: They propagate at the **speed of sound** in the material

Modes: There are typically **three acoustic modes**:

- **1 longitudinal acoustic (LA)** — atoms move along the wave direction
- **2 transverse acoustic (TA)** — atoms move perpendicular to the wave direction

Importance:

- Carry **heat** (thermal conductivity in insulators)
- Carry **sound** through solids

Optical phonons

Nature: Atoms in the **basis** move **out of phase** — one sublattice moves in one direction, while the other moves in opposite direction

Energy: Higher frequency (more energetic) (~60 meV in Si)

Existence: Only in crystals with **more than one atom per primitive cell** or **relative displacement of the atoms within the unit cell**

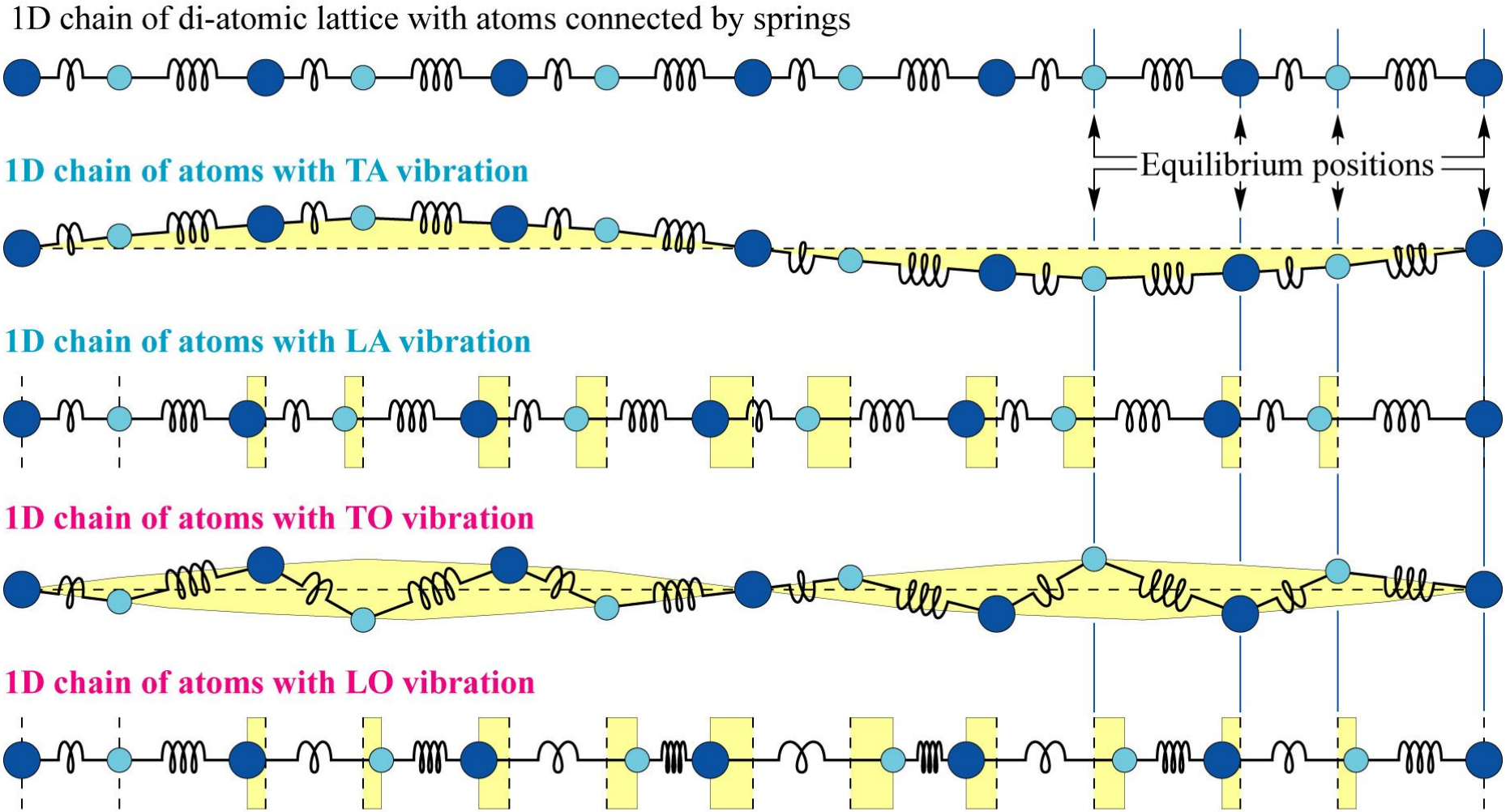
Interact with electromagnetic radiation, especially **infrared (IR)** light — hence the name *optical* phonons

Modes: 1 longitudinal optical (LO) and 2 transverse optical (TO)

Importance:

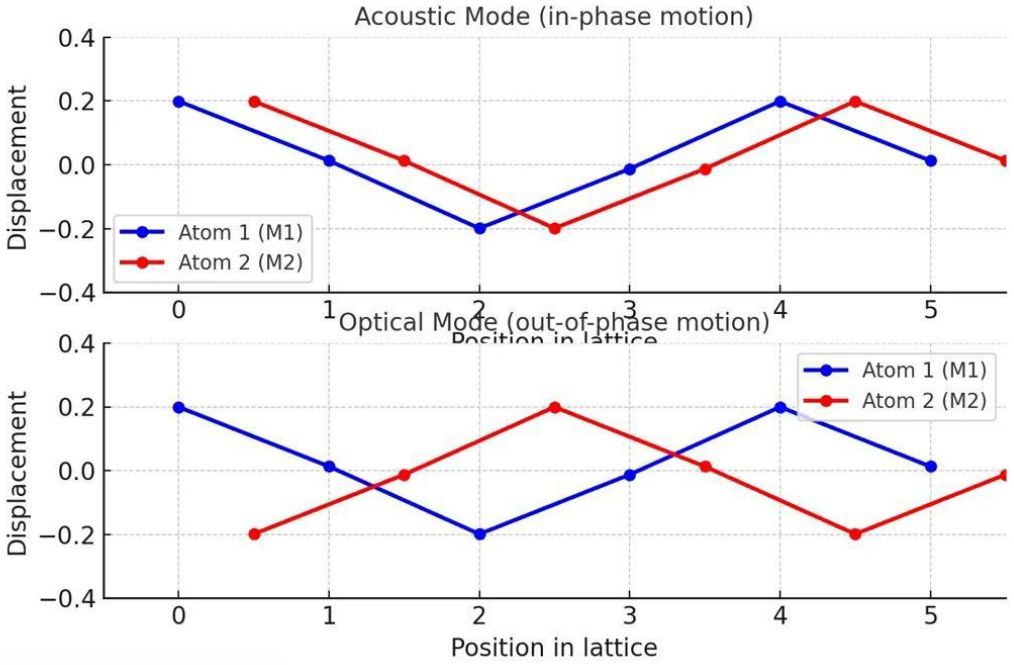
- Important in **infrared absorption**, **Raman scattering**, **electron-phonon interaction**
- Play a key role in **semiconductor device physics** (like carrier mobility, hot electron effects)

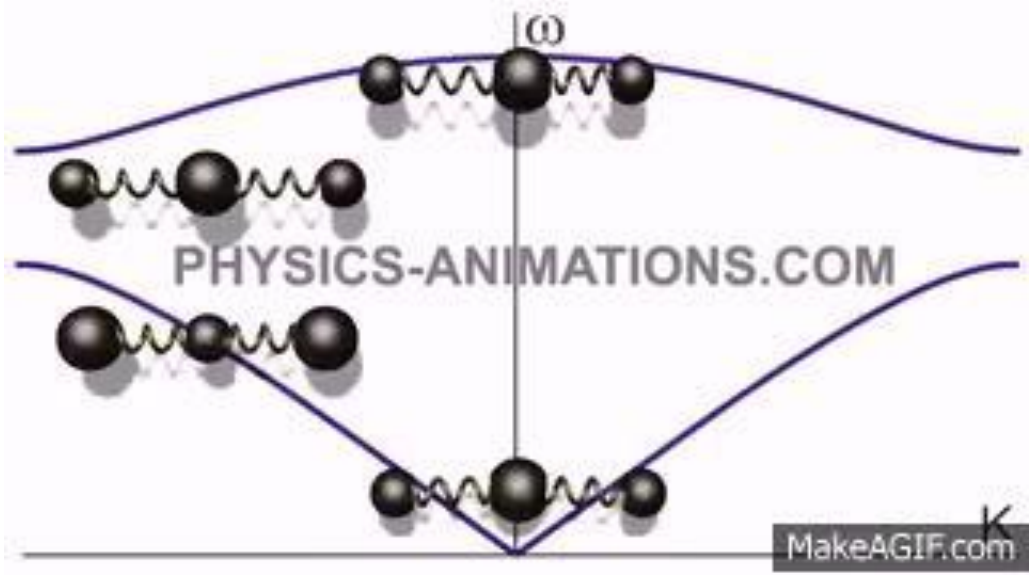
Vibrational models of a diatomic lattice



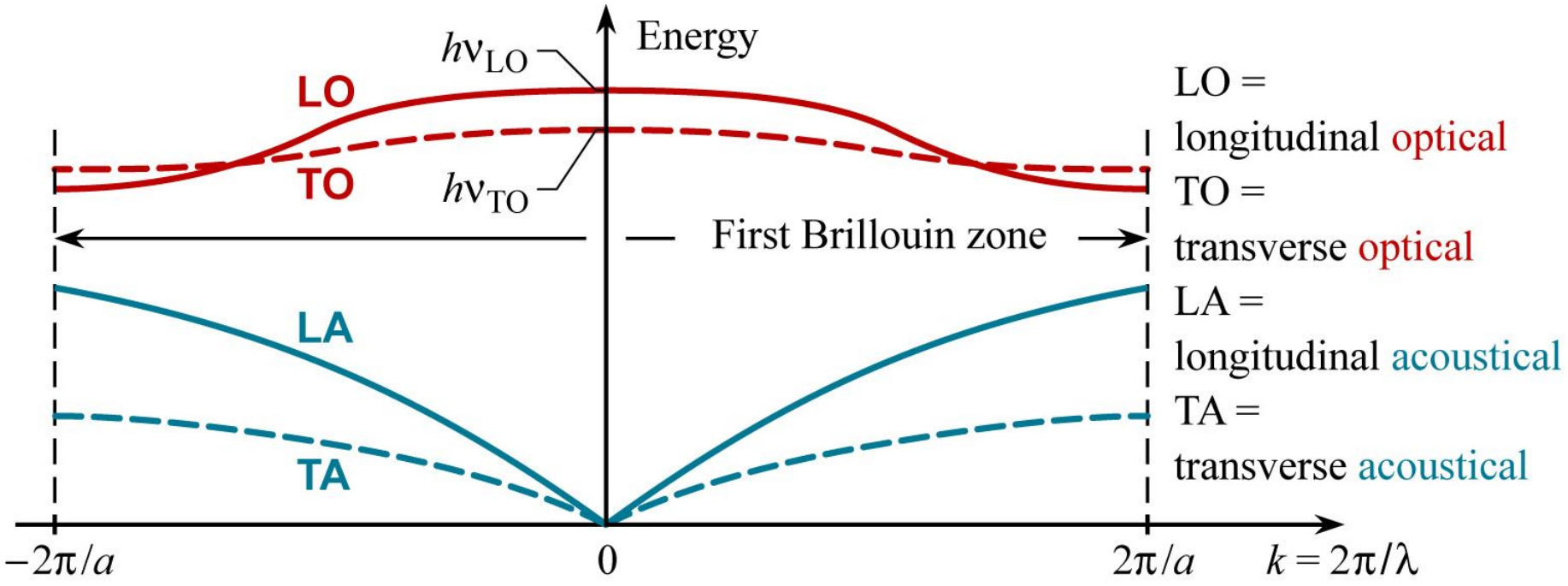
Adjacent atoms are displaced in the same (opposite) direction for acoustical (optical) phonons

Vibrational models and the dispersion relation of a diatomic lattice





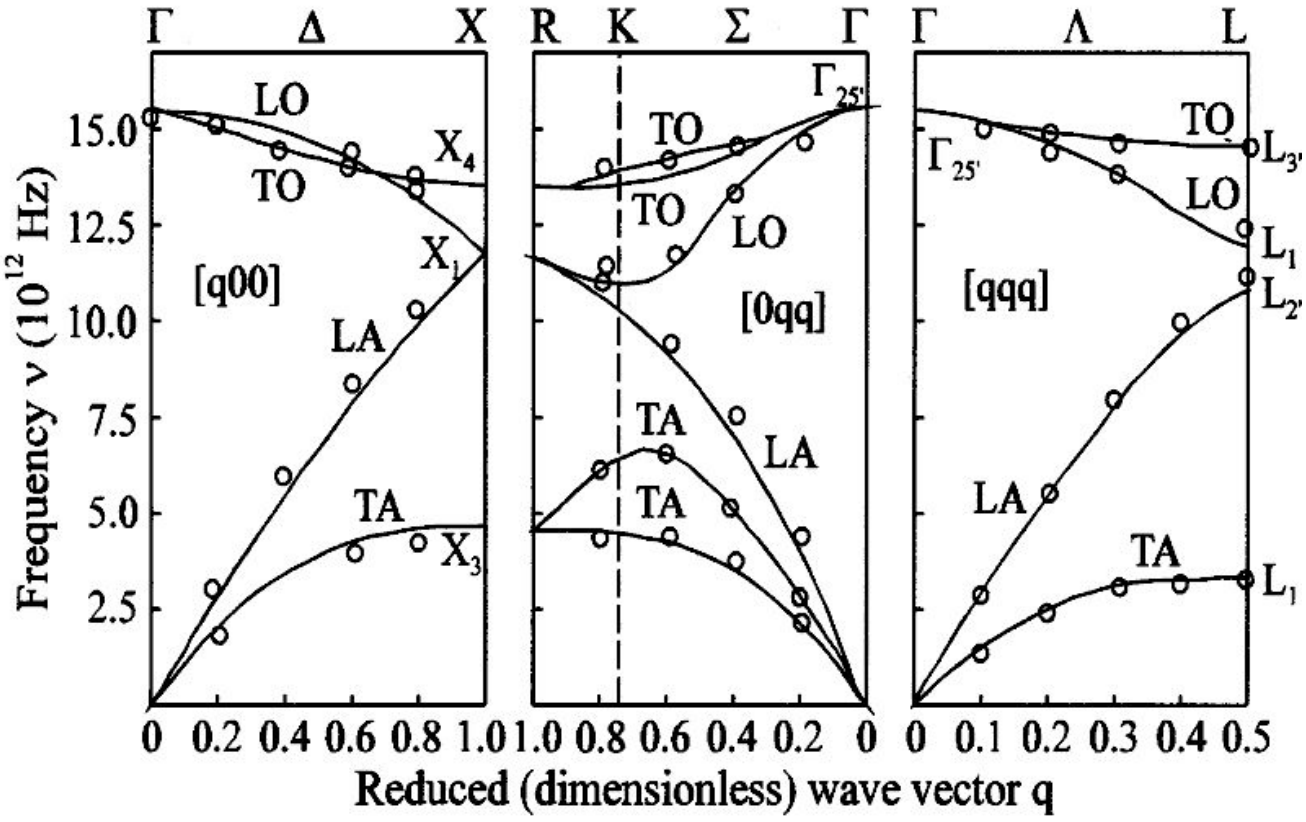
Phonon dispersion relation in a nutshell

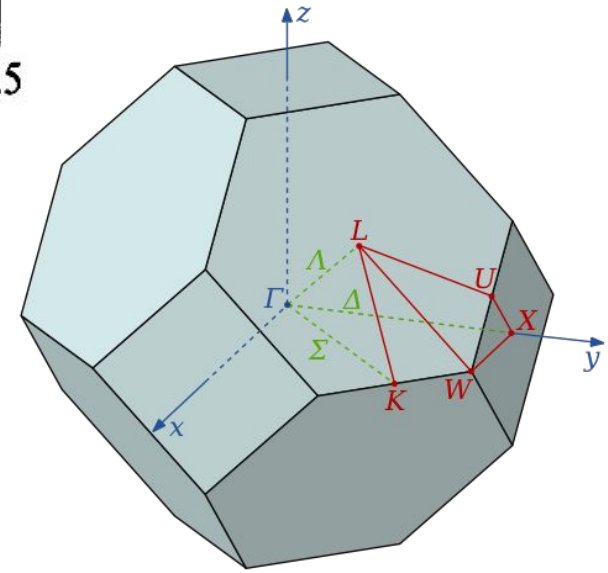


The dispersion relation can be derived by applying Newtonian mechanics to a mass-spring system

See here

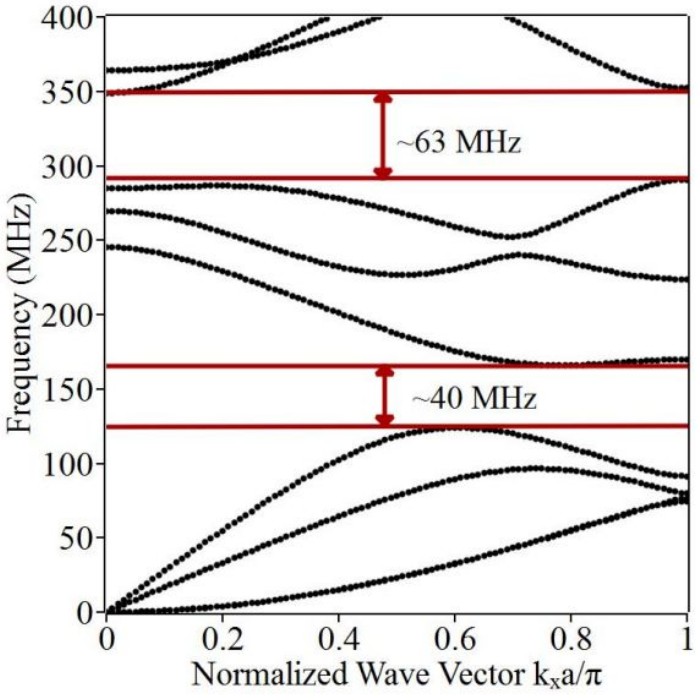
Phonon dispersion of bulk Silicon

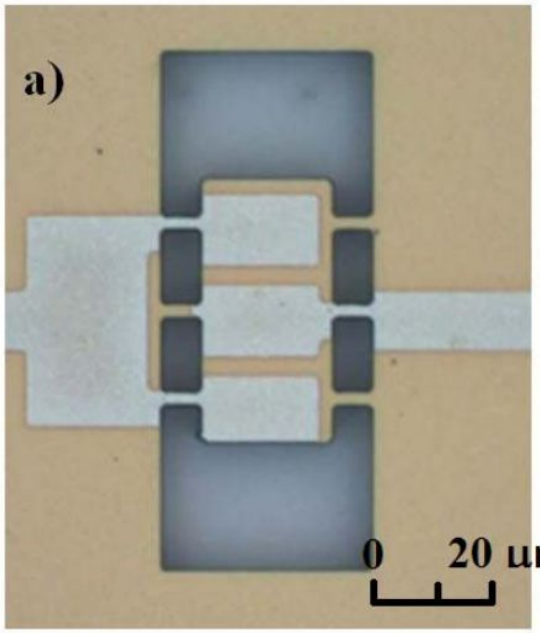


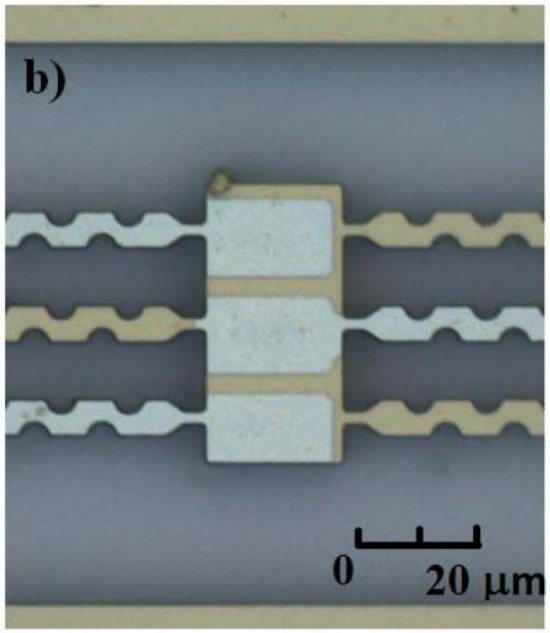


Phonons also show (like electrons) anisotropic dispersion relationship

Phononic dispersion may have bandgaps





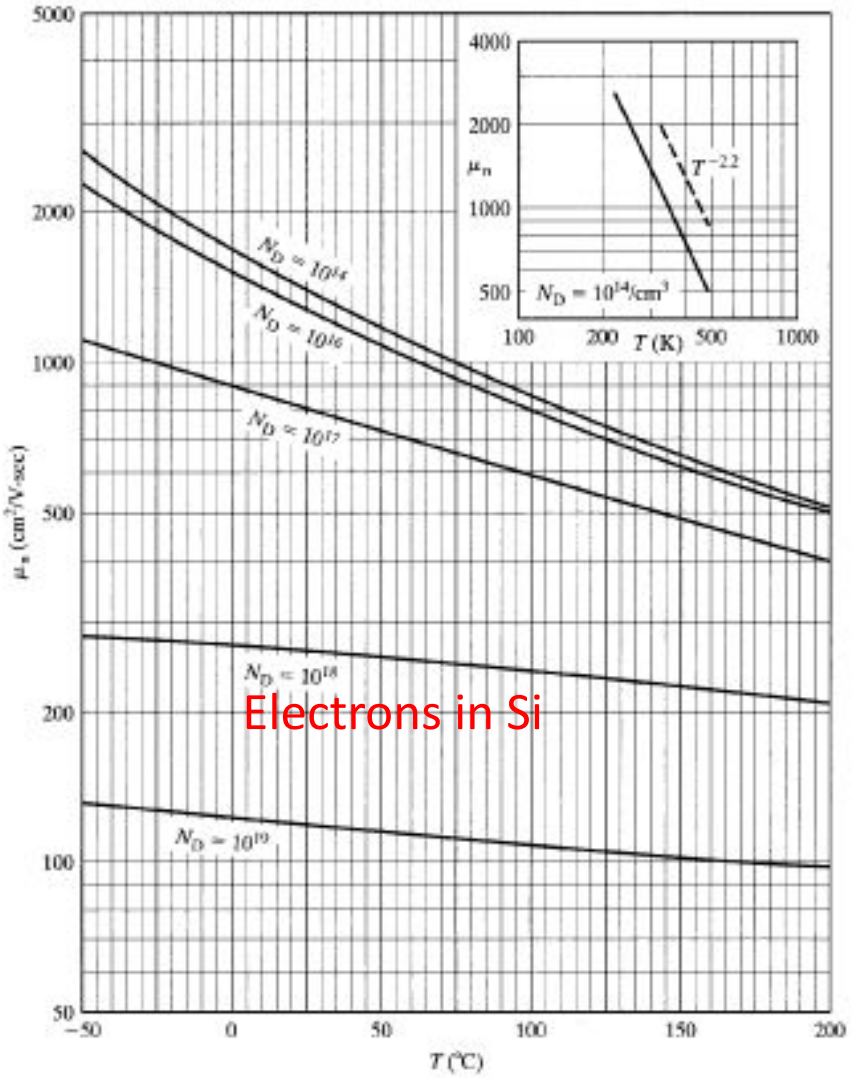


Phononic crystals - Engineered structures to get a desired dispersion relationship

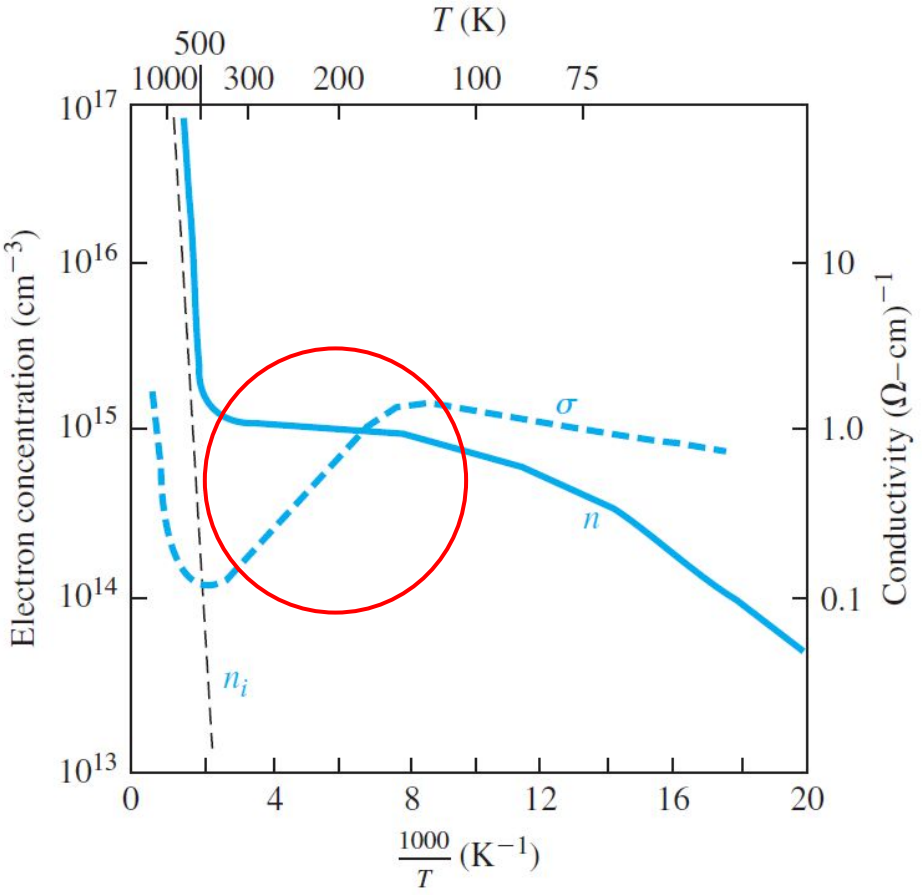
Anchors have a phononic bandgap at the resonance frequency to minimize energy loss through anchors. This improves the Quality factor of MEMS resonators

https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=7874145

Why TCR of intrinsic and extrinsic semiconductors have different polarity?

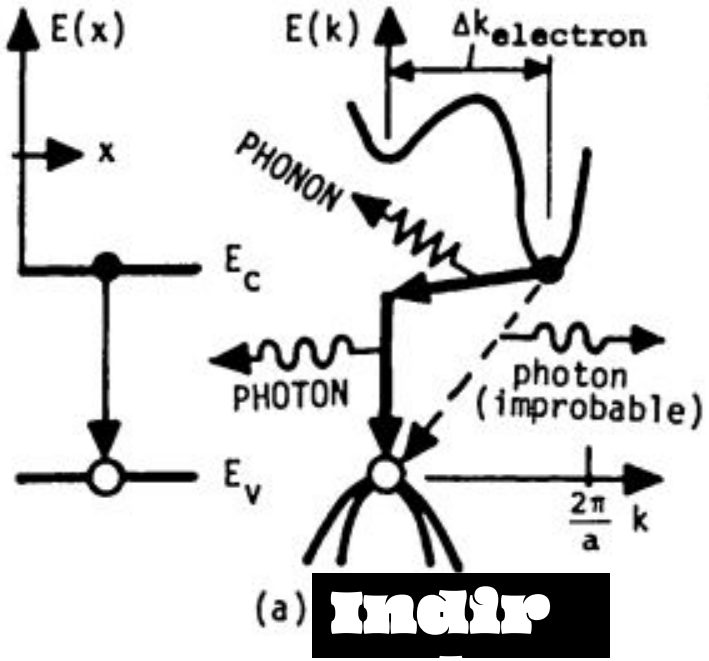


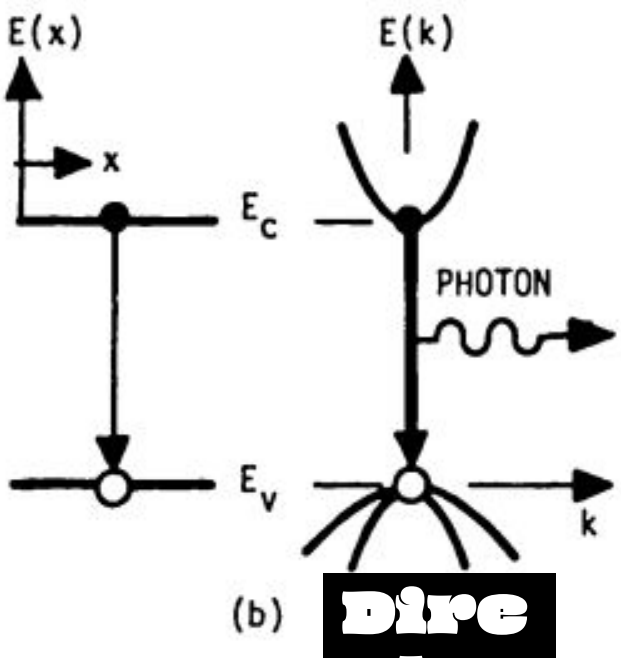
Increase in phonon scattering reduces mobility

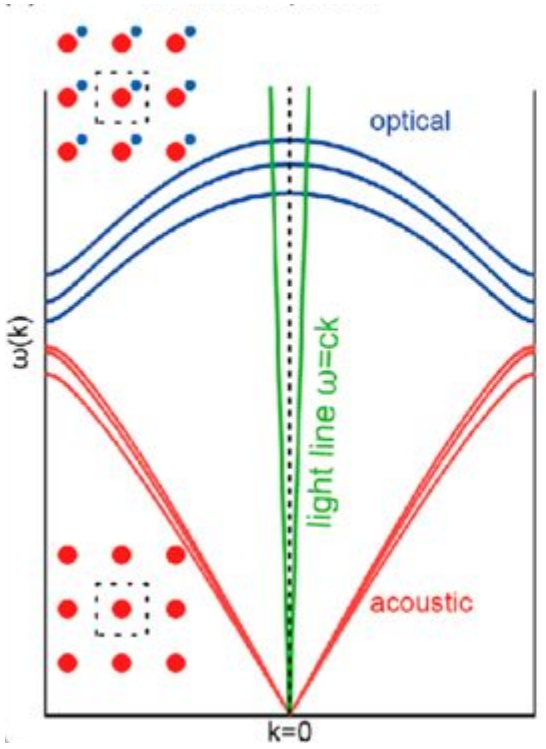


Extrinsic semiconductor has positive coefficient of resistance at room temperature

Phonons facilitate electron-hole recombination







$\lambda = hc/E = 1.24\text{eV}/E \quad \mu\text{m}$

$= 12400\text{eV}/E \quad \text{A.}$

$P_{photon} = h/\lambda = 4.1357\times10^{-15}(\text{eV-sec})/10^{-4}(\text{cm})$

$= 4\times10^{-11}\ \text{eV}/(\text{cm/sec}).$

$\Delta P_{electron} = \hbar\Delta k \propto \hbar(2\pi/a) = h/a$

$= 4.1357\times10^{-15}(\text{eV-sec})/10^{-8}(\text{cm})$

$= 4\times10^{-7}\ \text{eV}/(\text{cm/sec})$

Change of momentum is so large that it can't be carried by a photon. Hence light emission is very inefficient in indirect bandgap semiconductors